MASTER DOCUMENT

Critical Review of Agbola and Acupan (2010): An Empirical Analysis of International Labour Migration in the Philippines

Agbola, F. W., & Acupan, A. B. (2010). *Economic Systems*, *34*(4), 386–396.
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How to read this document

Each part pairs **highlighted excerpts from the article** (screenshots, page numbers in the captions) with a **concise technical review**. Colours carry meaning:

Key fact / result from the paper	Strength of the study	Critical note (weakness)	Reviewer suggestion

The article at a glance

Item	Summary
Question	How do economic, demographic and political factors affect emigration from the Philippines?
Setting	Philippines; 8.2 million Filipinos abroad; US$14.4 billion remittances (2007)
Data	Annual national time series, 1975–2005 (T = 31); CFO, NSCB, IMF IFS, BSP
Method	Unit root and cointegration tests → unrestricted error correction model (ECM) by OLS
Main result	Unemployment ↑ emigration; adult literacy, political instability, population density ↓ emigration
Verdict	Careful time-series steps, but small sample, weak proxies and no pull factors

Table 1. Overview of the study and of this review.

1. Research Problem / Gap

> **Guiding question**
> What issue or knowledge gap does the study address?

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ABSTRACT

This study empirically investigates the impact of economic, demographic, and political factors on the size of emigration from the Philippines. In 2007, overseas workers from the Philippines sent remittances in excess of US$14 billion annually to their families back home. Although these remittances are an important source of foreign exchange and play an important role in economic development, the determinants of emigration in the Philippines are not well established. A simple unrestricted error correction model of migration was specified and estimated using data spanning the period 1975–2005. Results indicate that the level of unemployment, adult literacy and population density are the key determinants of emigration in the Philippines. The result also indicates that government instability impacts negatively on emigration in the Philippines. The policy implications of the results are discussed.

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Excerpt 1.1. Abstract: the gap and the headline result (p. 386).

The objective of this paper is to empirically investigate the impact of economic, demographic and political factors on the size of emigration in the Philippines. A simple unrestricted error correction model of migration was specified and estimated using annual data spanning the period 1975–2005 in a time-series theoretic framework. The findings of this study provide an understanding of how the historical conditions, as shaped by socioeconomic and political systems, have reshaped international labour migration in the Philippines. This paper outlines how these findings could serve as a useful guide for government policymakers seeking to formulate and implement policies to address the issues concerning international labour migration in the Philippines.

Excerpt 1.2. Objective of the study (p. 387).

Section 2c states that:

> "...the State does not promote overseas employment as a means to sustain economic growth and achieve national development. The existence of the overseas employment program rests solely on the assurance that the dignity and fundamental human rights and freedoms of the Filipino citizen shall not, at any time, be compromised or violated."

Section 2i states that:

> "Nonetheless, the deployment of Filipino overseas workers, whether land-based or sea-based, by local service contractors and manning agencies employing them shall be encouraged".

Excerpt 1.3. The policy ambiguity in Republic Act No. 8042 (p. 388).

Review

- **Issue.** Labour export began in the 1970s as a stop-gap for unemployment and a balance-of-payments crisis, then became structural: 8.2 million Filipinos abroad and US$14.4 billion in remittances (2007).
- **Gap.** The determinants of emigration "are not well established" (p. 386); there is no country-specific, long-run time-series evidence on domestic drivers.

- **Policy tension.** RA 8042 §2c says the State "does not promote" overseas work, while §2i says deployment "shall be encouraged". A de facto labour-export policy lacks an evidence base.
- **Research question.** How do economic, demographic and political factors affect the size of emigration?

> **Strength**
> Clear, policy-relevant question for one of the world's largest labour exporters.

> **Critical note**
> The gap is **asserted, not demonstrated**: there is no systematic review of earlier Philippine migration studies.

2. Theoretical Framework / Theory

> **Guiding question**
> What theory or conceptual framework guides the study?

3. The model and data

A number of theories have been advanced in the economic literature to explain the factors that influence the decision to migrate. An economic theory of migration analyses the allocation of labour across international boundaries where the underlying premise is that individuals migrate because of tangible and intangible benefits. Individual decisions to migrate are, however, constrained by immigration policies and socio-economic and demographic factors of the individual. As Hoddinott (1994) emphasises, these explanations of migration based on behavioural decisions of the individual give primacy to Ravenstein's (1885) sixth law that individuals move in order to 'better' themselves. This concept is espoused in the migration model proposed by Todaro (1969), where it is argued that migration is the outcome of individual utility maximisation. Several authors have extended Todaro's migration model to include household utility maximisation. Contributions to this area of research include those by Connell et al. (1976), Rempel (1981), Stark and Levhari (1982), and Low (1986), among others. Hoddinott (1994) is critical of this approach, arguing that these theories of migration reflect an oversimplification of the behaviour of migrants. It also fails to explain the role prospective migrants play in the decision-making process. Hoddinott further argues that these migration models fail to provide an insight into how costs and benefits of migration are shared.

Clearly, the motivation to migrate may be described as influenced by a combination of socioeconomic, ethnic and political factors. Borjas (1989) provides a succinct review of migration theories, which are hence not discussed here. However, a brief summary of the three main theories identified by Borjas (1989), which have shaped recent methodological frameworks for modelling migration behaviour, is presented. The first is the human capital investment theory proposed by Hicks (1939) and espoused by Sjaastad (1962). This theory posits that migration rises the higher the returns, and vice versa. The second is the asymmetric information theory which is based on the concept of 'brain drain'. This theory posits that host countries have more information about immigration skills than the source country and are therefore capable of influencing migration patterns (see Kwok and Leland, 1982, for further discussion). The third theory, popularised by Piore (1979), is based on the conjecture that the impact of migration on natives in the receiving country is likely to be small because immigrants are forced to take jobs that natives refuse to accept. Borjas is critical of these theories though, arguing that they fail to provide a sound theoretical foundation for explaining three important questions relating to migration, namely, the factors that determine the direction, size and composition

Excerpt 2.1. Migration theories reviewed by the authors (p. 389).

Clearly, there is currently no single unified theory of emigration that addresses these issues. Instead, a number of theories and hypotheses continue to emerge aimed at examining the determinants of migration. The migration model is a relationship linking migration to variables identified by economic theory. In accordance with the existing literature, a migration model is developed for the Philippines, starting from a gravity model, which is a reduced form equation derived from a system of demand and supply relationships, and assuming that supply and demand are influenced by income, which captures the capacity to migrate, population and factor endowments. This study adopts a macroeconomic perspective and focuses on domestic factors impacting on international migration in the Philippines. Following the extant literature, particularly the studies by Greenwood (1975), Borjas (1987, 1989), and Adams and Page (2003), the Philippines migration model is specified as follows:

$$m = \alpha_0 + \alpha_1\, cm_t + \alpha_2\, adl_t + \alpha_3\, ineq_t + \alpha_4\, ineqsq_t + \alpha_5\, govs_t + \alpha_6\, ir_t + \alpha_7\, pd_t + \alpha_8\, ur_t + \alpha_9\, y_t + \varepsilon_t \quad (1)$$

where the dependent variable m is the share of migrants in total Philippines population, *cm* is the cost of migrating from the Philippines to various labour-receiving countries, *adl* is the adult literacy rate, *ineq* is the level of income inequality as measured by the gini coefficient, ineqsq is the level of income inequality-squared, *govs* is the measure of political instability, *ir* is the inflation rate, *pd* is the population density as measured by the number of people per square kilometre, *ur* is the unemployment rate, *y* is the real per capita income. This study employs the general to specific approach of Hendry and Ericsson (1991) in the specification of the dynamic migration model with the insignificant coefficients eliminated sequentially and reparametised to obtain a parsimonious specification. Hence, the final model was estimated in first difference with the error correction term, which captures the long-run time-series properties of the data series.

Excerpt 2.2. No unified theory; the gravity-based model, equation (1) (p. 390).

Review

- **Expected-utility models** (Ravenstein, 1885; Todaro, 1969): people migrate when the expected gain abroad exceeds the cost.
- **Human capital theory** (Hicks, 1939; Sjaastad, 1962): migration is an investment, so education should raise emigration.
- **Roy effect / self-selection** (Borjas, 1987): skill-return differentials decide who leaves; tested with the Gini coefficient and its square.
- **Gravity model** (Greenwood, 1975; Adams & Page, 2003): a reduced-form demand–supply relation in income, population and migration cost.
- **Operational model, equation (1):** m = f(*cm*, *adl*, *ineq*, *ineq*², *govs*, *ir*, *pd*, *ur*, *y*), a macro push-factor equation for the migration share.

All theory citations are as cited in Agbola and Acupan (2010).

Strength

Grounded in established migration theory; each hypothesis gives a testable expected sign.

Critical note

Eclectic, not structural: the theories mainly justify coefficient signs.

The gravity model is bilateral by nature, yet no destination-side (pull) variables enter the model.

3. Data

Guiding question

What data were used, and where were they obtained?

The data on migration were obtained from the Commission on Filipinos Overseas (CFO) and the National Statistics Coordinating Board (NSCB) in the Philippines (NSCB, 2006), and include both temporary labour workers and permanent Filipino migrants living and working overseas. Data on population density were obtained from NSCB (NSCB, 2006). Data on Philippines gross domestic product, unemployment rate, and consumer price index were obtained from the International Monetary Fund's (IMF) *International Financial Statistics* CD Rom (IMF, 2006b) and supplemented with official records of the *Bangko Sentral ng Pilipinas* (BSP) (BSP, 2006) and the NSCB (NSCB, 2006). Data on adult literacy rate were obtained from BSP (BSP, 2006) and the NSCB (NSCB, 2006).

Excerpt 3.1. Data sources (p. 390).

Data problems continue to constrain studies on migration and this study is no different. These data problems relate to the way in which data related to international migration are collected and compiled. It is therefore important to spell out how the data were gathered. The data used in this study are annual data from 1975 to 2005. The cost of migration was estimated as the average cost of a one-way airline ticket from the capital of the Philippines to the top 24 country-destinations of Filipino workers. It is important to note that the use of one-way airline tickets may underestimate the real costs of migration, given that it does not take into account other costs of migration, such as the cost of job placement and the cost of travelling from the place of residence in the Philippines to the country's

Excerpt 3.2. Sample period and the airfare proxy for migration cost (p. 391).

capital, as well as administrative costs, which can be enormous (Taylor, 2006). However, due to the lack of reliable data on the other costs of migration, this study uses the one-way airline ticket as the proxy for the cost of migration. The proxy used to represent the cost of migration only represents, at best, the minimum costs associated with migration. Hence, the impact of this variable on migration should be interpreted with caution. It is hypothesised that a negative relationship exists between the cost of migration and the number of migrants.

Based on human capital theory, the more educated a person is, the more likely it is that they will migrate. This study aims to investigate this hypothesis for the Philippines. Data on the number of people with primary or secondary education are not readily available, particularly for the study period. The adult literacy rate is therefore used as a proxy for the level of education of Filipinos. This study hypothesises a positive relationship between adult literacy rate and migration. Previous studies have found the income of migrants to impact significantly on migration. This study incorporates income measured as per capita gross domestic product in purchasing power parity terms. A negative relationship is hypothesised between income and migration. This study also incorporates inequality, measured as a Gini coefficient and inequality-squared to capture the Roy effect in the migration literature. The Roy effect is based on the premise that differences in the return to skills will select migrants from different parts of the skill distribution (Clark et al., 2003). The inflation rate and unemployment rate variables are included in the model to capture the effects of domestic economic activity on migration flow. A high inflation rate, which is indicative of economic malaise, is expected to impact negatively on migration flow. A high unemployment rate is expected to exert an upward pressure on migration flow as people migrate in search of jobs. This study includes the policy variable of government stability as measured, which was derived as the frequency of change in government leadership and the number of attempted and successful uprisings against the government.

Excerpt 3.3. Proxies and expected signs (p. 392).

Variable	Measure / proxy	Source
Migration share (*m*)	(Temporary + permanent migrants) ÷ population	CFO; NSCB
Cost of migration	One-way airfare, Manila → top 24 destinations (proxy)	Authors' estimate
Adult literacy	Adult literacy rate (proxy for education)	BSP; NSCB
Income inequality, ineq²	Gini coefficient and its square	Not stated
Political instability	Leadership changes, coups and uprisings	Gastil (1987) approach
Inflation; unemployment; income	CPI inflation; unemployment rate; real GDP per capita (PPP)	IMF IFS; BSP; NSCB
Population density	People per km²	NSCB

Table 2. Variables, measures and sources (adapted from Agbola & Acupan, 2010, pp. 390–392).

- **Sample.** Annual national aggregates, 1975–2005: T = 31, all secondary data.

Critical note

- **Undisclosed data handling:** the Gini source is not stated; the Philippine Gini is published only every few years, so the annual values were presumably interpolated.
- **Aggregation:** *m* pools temporary and permanent migrants, who respond to different incentives.
- **Weak proxies:** airfare is a lower bound on migration cost; near-universal adult literacy has little variation.

4. Methodology

Guiding question

What research methods and analytical approaches were employed?

> This study utilises the Augmented Dickey–Fuller (ADF) test (Dickey and Fuller, 1979) and Phillip–Perron (P–P) (Phillips and Perron, 1988) unit root tests to investigate the time-series properties of the data series. Table 1 reports the unit root test results. From Table 1, except for the P–P test for the income variable, the ADF and P–P test results indicate that all the data series fail to reject the null hypothesis of a unit root in levels, but become stationary in first difference. Given that all the data series are of the same order of integration for the ADF tests, which is *I*(1), we proceed to perform the Johansen cointegration test.
>
> We start by performing a preliminary test for cointegration based on the Engel–Granger test (Engle and Granger, 1987). The results led to the rejection of the null hypothesis of no cointegrated series and

Excerpt 4.1. Unit root tests and the move to cointegration (p. 390).

Table 1
Unit root test results.

Variable	Augmented Dickey–Fuller test		Phillip–Perron test	
	Level	First difference	Level	First difference
m	−2.437	−3.838	−2.321	−5.776
cm	−1.668	−4.010	−0.264	−9.900
adl	−0.968	−5.523	−1.196	−5.723
ineq	−1.992	−7.232	−2.298	−4.562
ineqsq	−1.513	−7.291	−1.742	−4.439
govs	−1.712	−6.762	−1.853	−6.657
ir	−3.030	−6.756	−3.028	−16.900
pd	−1.503	−8.127	−1.309	−13.439
ur	−2.711	−6.809	−2.698	−7.514
y	−0.626	−5.369	−2.357	−2.123

Note: The McKinnon (1991) critical values are −2.623 for 10%, −2.967 for 5% and −3.679 for 1% level.

the acceptance of the existence of at most a single cointegrated vector given that the *t*-statistic of −4.485 of the coefficient of the lagged residual in the test equation is greater than the critical values of −3.98, −3.42 and −3.13 at the 1%, 5% and 10% levels, respectively. However, as Agbola and Damoense (2005) note, the Engel–Granger approach assumes the existence of at most a single cointegration vector and is also sensitive to the choice of dependent variable. To overcome these limitations, we proceed to determine the presence of an equilibrium-type relationship between the variables using the Johansen's multivariate test (Johansen and Juselius, 1990). Table 2 reports the Maximal Eigenvalue and Trace tests results for cointegration. The results of the Maximal Eigenvalue and Trace tests indicate that we can reject the null hypothesis of no cointegration. The Maximal Eigenvalue test indicates five cointegrated vectors, while the Trace test indicates that there are eight cointegrated vectors at a 5% significance level. Furthermore, we perform the bounds test approach suggested by Pesaran et al. (2001). This technique is superior to the Engel and Granger and Johansen cointegration tests because it can be performed regardless of whether the underlying order of integration of the data series are of the same order or not. For the Pesaran bounds test, we reject the null in favour of the alternative hypothesis given that our sample test statistic exceeds the relevant upper critical value. We conclude that there exists a long-run relationship between all the variables in the Philippine migration model.

Excerpt 4.2. Table 1 (unit roots) and the cointegration evidence (p. 391).

Review

A quantitative, single-country **time-series econometric** design in five steps:

- **Step 1 – Unit roots:** ADF and Phillips–Perron tests; the series are treated as I(1).
- **Step 2 – Cointegration:** Engle–Granger ($t = -4.485$), Johansen (λmax: 5 vectors; trace: 8) and the Pesaran bounds test all confirm a long-run relationship.
- **Step 3 – Specification:** Hendry general-to-specific reduction to a parsimonious model.
- **Step 4 – Estimation:** unrestricted ECM by OLS (EViews 6.0).
- **Step 5 – Diagnostics:** Breusch–Godfrey, RESET, Jarque–Bera, ARCH and CUSUM/CUSUMSQ, all passed.

> **Strength**
> Textbook sequence with three cointegration tests and a full set of diagnostics.

Critical note

- One ECT despite 5–8 cointegrating vectors: the long-run structure is collapsed for parsimony.
- **Order of integration unclear:** income (*y*) fails the P–P test even in first differences (−2.123 vs. the 10% critical value of −2.623).
- **Low power:** T = 31 for 10 estimated terms, and there are no structural-break tests (1986 EDSA, 1997 Asian financial crisis).

5. Key Findings

Guiding question

What are the main results and conclusions of the study?

4. Results and discussion

Table 3 reports a simple unrestricted error correction model of migration for the Philippines by ordinary least squares method in the Eviews 6.0 econometric package. While the Johansen test suggested the existence of more than one cointegrating vector, the general to specific approach favoured a more parsimonious model with one error correction term and tests for the adequacy of this parsimonious model supports this. The error correction term (ECT_{t-1}), which accounts for the short-run adjustment of the cointegrated variables, is negative and estimated to be −0.77, indicating a convergence towards a long-run steady state in a little over a year. Most of the estimated parameters

Table 3
Unrestricted ECM of migration for the Philippines.

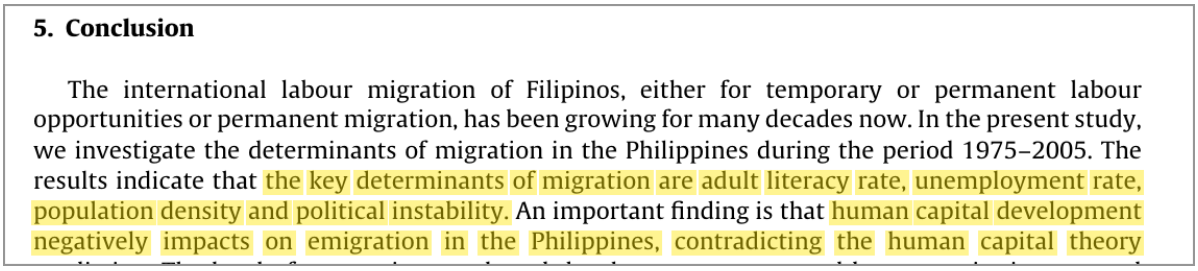
Variable	Coefficient	*t*-Statistic	Prob.
Constant	1.6350	3.573	0.020
Δcm	−0.0017	−0.429	0.673
Δadl	−0.6590	−5.225	0.000
$\Delta ineq$	−171.195	−1.594	0.127
$\Delta ineqsq$	165.927	1.323	0.202
$\Delta govs$	−0.4820	−3.444	0.003
Δir	−0.0358	−1.327	0.200
Δur	0.2361	4.629	0.000
Δpd	−0.2541	−2.756	0.013
Δy	−0.0002	−0.293	0.773
ECT_{t-1}	−0.776	−4.224	0.000
Diagnostics			
R^2-adjusted = 0.63			
F-value = 5.848 (p = 0.000)			
Breusch–Godfrey serial correlation LM test: χ^2 = 1.099 (p = 0.294)			
Ramsey RESET mis-specification test: χ^2 (1) = 0.010 (p = 0.919)			
Jarque–Bera normality test: χ^2 (1) = 4.962 (p = 0.084)			
ARCH test: χ^2 (1) = 0.312 (p = 0.577)			

Excerpt 5.1. Table 3, unrestricted ECM; the significant rows are highlighted (p. 392).

Explanatory variable	Coefficient	t-statistic	Result (5%)
Unemployment	+0.236	4.629	Significant ↑
Adult literacy	−0.659	−5.225	Significant ↓
Political instability	−0.482	−3.444	Significant ↓
Population density	−0.254	−2.756	Significant ↓
Inequality; inequality²	−171.195; +165.927	−1.594; 1.323	Not significant
Inflation; cost of migration; income	−0.036; −0.002; −0.0002	−1.327; −0.429; −0.293	Not significant
Error correction term (ECT)	−0.776	−4.224	Significant

Table 3. ECM results. Adjusted $R^2 = 0.63$; $F = 5.848$ ($p = 0.000$). Adapted from Agbola and Acupan (2010, p. 392).

- **Fit and adjustment:** the model explains 63% of the variation in the migration share; ECT = −0.776 means about 78% of any deviation from equilibrium is corrected within a year.
- **Unemployment ↑ → emigration ↑:** emigration acts as a safety valve for surplus labour.
- **Adult literacy ↑ → emigration ↓:** this contradicts human capital theory; the authors point to the low-skilled profile of migrants.
- **Political instability ↑ → emigration ↓:** emigration fell in turbulent years (1983, 1986, 2001, 2005).
- **Population density ↑ → emigration ↓:** contrary to expectation; the authors suggest, tentatively, that domestic demand creates local jobs.
- **Not significant:** income, inflation, migration cost and inequality, so the Roy effect is rejected.



5. Conclusion

The international labour migration of Filipinos, either for temporary or permanent labour opportunities or permanent migration, has been growing for many decades now. In the present study, we investigate the determinants of migration in the Philippines during the period 1975–2005. The results indicate that the key determinants of migration are adult literacy rate, unemployment rate, population density and political instability. An important finding is that human capital development negatively impacts on emigration in the Philippines, contradicting the human capital theory

Excerpt 5.2. Conclusion: the key determinants (p. 394).

domestic activity as captured by inflation rate were found not to impact on emigration in the Philippines. Another important finding is that the Roy effect does not hold for the Philippines.

In the last decades or so, international labour migration has emerged as one of the main factors in social transformation and development in the Philippines. Its significance looks set to grow as the factors affecting the flow of international labour migration change in response to changing economic conditions domestically and abroad. A review of migration policy reforms in the Philippines suggests the lack of effective monitoring of international migration in the Philippines and the extent to which economic factors influence the decision of migrants. This is due in part to the failure on the part of the government to acknowledge migration as a developmental strategy and make a concerted effort to ensure the smooth and efficient flow of migrants. The results of this study underscore the importance of the need for the Philippine government to pursue policies of ensuring effective monitoring and collection of comprehensive statistics on international migration in order to assess the impact of changing economic and political conditions on migration in the Philippines. The recent establishment of bilateral agreements by the Philippine government with migrant receiving countries on the regulation of migration and the rights and condition of Filipino migrants could help to enhance protection and greater social security of Philippine citizens working abroad. In the end, the Philippine government should look into implementing policies and programs that will develop a more viable and vibrant domestic economy that is capable of absorbing the labour force in the Philippines and avoid the reliance on the unpredictable demand by migrant receiving countries.

Excerpt 5.3. Conclusion: the Roy effect and the policy implications (p. 395).

- **Policy conclusion:** better migration statistics, bilateral agreements to protect overseas workers, and a domestic economy able to absorb labour.

Critical note

- The **low-skill explanation conflicts with the paper's own source**: Alburo and Abella (2002, as cited in Agbola & Acupan, 2010) found professional outflows exceeding domestic additions.
- The insignificant results may partly reflect **low statistical power**, not true zero effects.

6. Areas for Further Research

Guiding question

What limitations were identified, and what future research opportunities are suggested?

The *population density* (*pd*) variable is found to be negative and statistically significant in influencing the migration flow in the Philippines. The negative impact of the population density variable on migration suggests that as the population increases in the Philippines the migration flow falls. This is contrary to expectations, as an increase in population is expected to result in an increased flow of migrants. For the Philippines government, this finding is important, as it demonstrates that there is an inherent tendency of the economy to discourage out-migration as the population grows. While this is difficult to rationalise, a possible explanation is that an increase in population will be associated with an increase in the demand for goods and services produced domestically. This will stimulate growth within the economy. As the welfare of individuals within the economy improves, they are unlikely to migrate. Further research investigating the reasons for this result is, however, warranted.

Excerpt 6.1. The authors call for further research on population density (p. 393).

Quite important in developing countries, such as the Philippines, is the political climate and the regularity of transfer of power in the country. This study follows the approach proposed by Gastil (1987), whereby a dummy variable is used as a measure to rate the political and civil liberty within the economy, with a higher rating indicating less freedom. In this study, the number of coups and coup attempts is used as a proxy for political instability. The coefficient of the *political instability* (*govs*) variable is negative and statistically significant at a 5% level, as expected. The result indicates that increased political instability reduces emigration from the Philippines. This finding is consistent with those of Karemera et al. (2000), who also found that increased political instability, which is often accompanied by increased restrictions on domestic travel, reduced international migration to North America. However, it is important to note that for the Philippines, there was no official restriction on residents' overseas travel even during the politically unstable years. The result indicates that as political stability deteriorates the level of migration declines. While our data does not permit us to move beyond speculation, the result appears consistent with events of the day. During the highly politically unstable years of 1983, 1986, 2001, 2005 – when there was growing restiveness and clamour amongst the general populace for a change in government – migration as a percentage of the population declined markedly. The periods immediately following these tumultuous times caused migration to rise, reflecting a renewed sense of national pride and hope (Spaeth, 2006). While other factors may also have influenced the decline in migration in those years, the result of this study adds some credence, at the very least, to the negative impact of political instability on the migration flow in the Philippines.

Excerpt 6.2. The authors acknowledge the limits of the political-instability evidence (p. 394).

Limitations identified by the authors

- Migration statistics are hard to collect and compile (p. 391).
- Airfare captures only the **minimum** cost of migration, so its coefficient should be read with caution (p. 392).
- The population-density result is unexplained: "further research … is warranted" (p. 393).
- The data "does not permit us to move beyond speculation" on political instability (p. 394).

Future research opportunities

- **Suggested by the authors:** explain the density effect; build systematic monitoring of migration statistics.

Reviewer suggestions

1. **Disaggregate** the migration share: temporary vs. permanent, and by skill or occupation.
2. **Add pull factors:** a bilateral panel gravity model with destination wages and labour demand (e.g., Gulf oil revenues).
3. **Extend past 2005 and test structural breaks** (1997, 2008, COVID-19) with Zivot–Andrews or Bai–Perron tests.
4. **Use household micro-data** to test human-capital selection directly.

Key Insights

> **Five takeaways**
>
> 1. **Push factors at home matter:** unemployment is the clearest driver of Filipino emigration.
> 2. **Education keeps people home** at the macro level, against human capital theory, but the explanation is contested.
> 3. **Instability and density reduce emigration**, and both mechanisms remain speculative.
> 4. **Long-run equilibrium:** about 78% of shocks to the migration share fade within a year.
> 5. **Read with care:** T = 31, weak proxies and missing pull factors limit the conclusions.

Bottom line. Agbola and Acupan (2010) offer useful early evidence that domestic push factors shape Filipino emigration. Without destination-side pull factors and richer data, however, they are not the whole story.

Reference

Agbola, F. W., & Acupan, A. B. (2010). An empirical analysis of international labour migration in the Philippines. *Economic Systems, 34*(4), 386–396. https://doi.org/10.1016/j.ecosys.2010.03.002